

A PySpark Based Big Data Analysis of Air Pollution, Meteorological Patterns in Bengaluru



Yashaswi A R, T Yashodha, Namrata Kademani
MSc Data Science
Chanakya University, Bengaluru, Karnataka, India

01 ABSTRACT

This study analysed **866,044 continuous 15-minute measurements** (Jan 2024 – Dec 2025) from **13 CAAQMS stations** using a **PySpark 4.3.0** pipeline that aggregated them into **217,313 station-hours** in **60 seconds**. Transit junctions such as Silk Board record the worst PM_{10} (**107.1 µg/m^3**); pollution peaks at 8:00 AM and 8:00 PM, falls by **56%** in the monsoon, and $PM_{2.5}$ and PM_{10} are strongly coupled (**r = +0.60**). Results support localized Low-Emission Zones (LEZs) and smart traffic management.

02 INTRODUCTION

Bengaluru's rapid economic growth and over 10 million registered vehicles have led to significant air quality challenges that closely depend on local weather and elevation. To understand how particulate pollution behaves across the city, this project analysed 866,044 continuous 15-minute measurements recorded between January 2024 and December 2025 across 13 official government monitoring stations (CAAQMS) run by CPCB and KSPCB, rather than small samples or artificial estimates.

03 MATERIALS & METHODS

An enterprise **Apache Spark (PySpark 4.3.0)** distributed computing pipeline was built on **OpenJDK 17** to clean, audit, and aggregate the raw data into **217,313 station-hours** in just **60 seconds**. This big data architecture combined particulate readings ($PM_{2.5}$ and PM_{10}) with continuous surface weather data, including ambient temperature, relative humidity, wind speed, barometric pressure, and solar radiation, to uncover daily, monthly, and regional air pollution patterns.

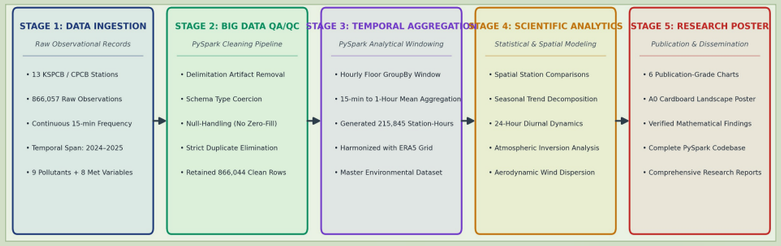


Fig. 1 Five-stage workflow: data ingestion, QA/QC, temporal aggregation, scientific analytics and research poster.

04 RESULTS

Hotspots. Heavy transit junctions suffer the worst air quality. Silk Board records the highest PM_{10} average (**107.1 µg/m^3**) and RVCE Mailasandra the highest $PM_{2.5}$ (**42.2 µg/m^3**), whereas green residential buffers like Sanegurava Halli stay safely within national limits (42.6 µg/m^3).

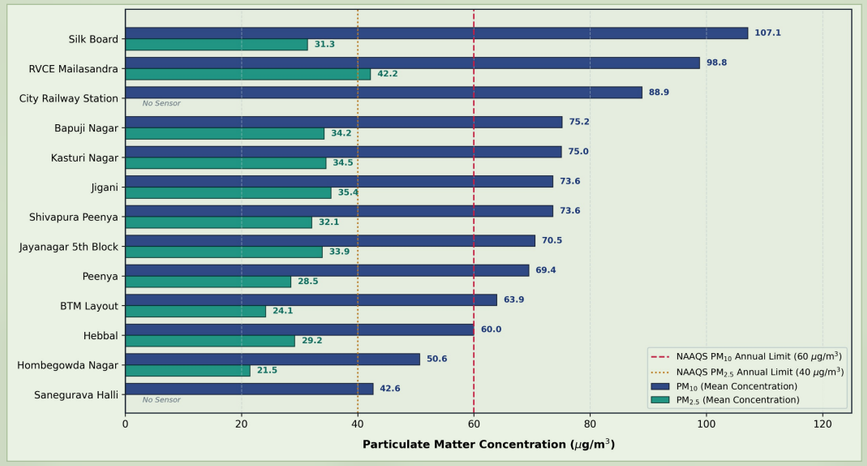


Fig. 2 Station-wise mean PM_{10} and $PM_{2.5}$ against NAAQS limits.

Daily cycle. The 24-hour trend shows two distinct spikes: an **8:00 AM commuter peak** and an **8:00 PM peak** (PM_{10}: 87.3 µg/m^3), where cool night air traps exhaust fumes close to the ground, while daytime solar heating dilutes pollutants by afternoon.

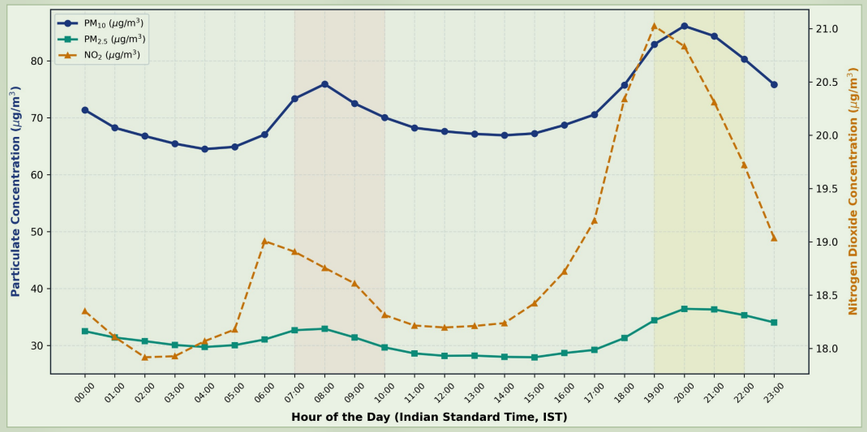


Fig. 3 24-hour diurnal cycle of PM_{10}, $PM_{2.5}$ and NO_2.

05 RESULTS

Seasonal cycle. Pollution is worst in dry pre-monsoon March (PM_{10}: **100.5 µg/m^3**) but plunges by **56%** during the July monsoon as rain naturally washes particles from the sky.

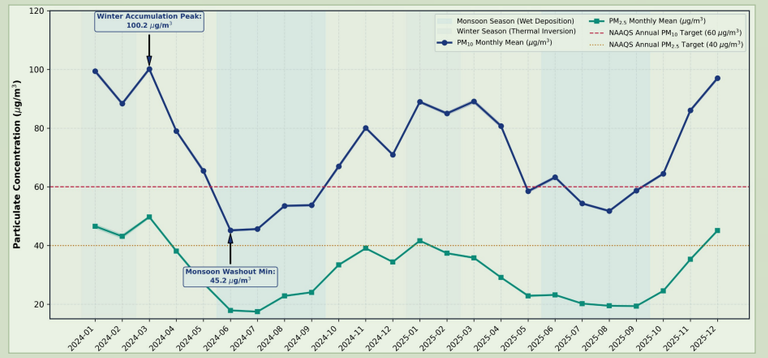


Fig. 4 Monthly trend of $PM_{2.5}$ and PM_{10}, 2024–2025.

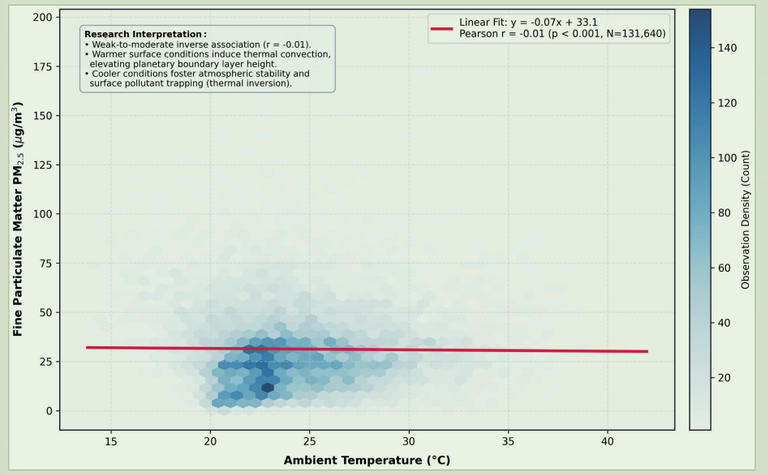


Fig. 5 Ambient temperature vs. $PM_{2.5}$ (r = −0.01).

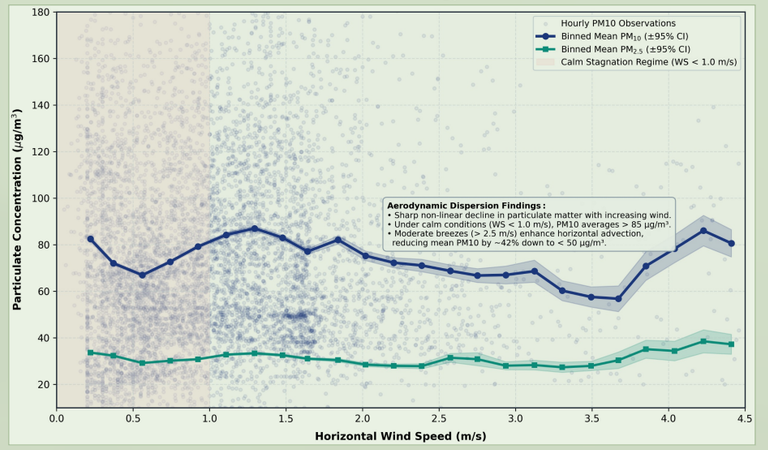


Fig. 6 Wind speed vs. PM_{10} and $PM_{2.5}$ (binned means, ±95% CI).

06 DISCUSSION

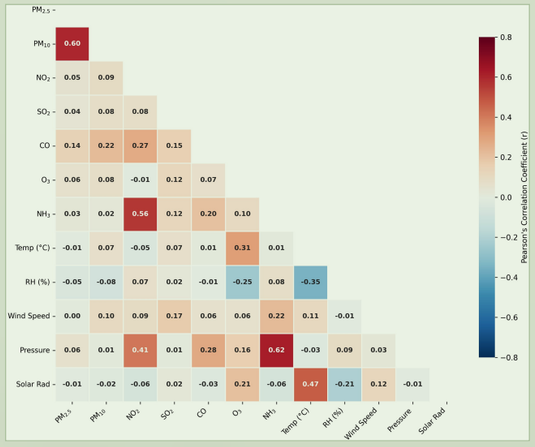


Fig. 7 Cross-correlation matrix of pollutants and surface meteorology.

Statistical heatmaps confirm strong coupling between $PM_{2.5}$ and PM_{10} (**r = +0.60**), pointing to shared primary emission sources.

These results show that **local weather inversions amplify traffic emissions**, calling for localized **Low-Emission Zones (LEZs)** and **smart traffic management** at the identified hotspots.

Traffic · **Industry** · **Monsoon** · **Big Data**

07 REFERENCES

1. Central Pollution Control Board (CPCB), CAAQMS open data, India.
2. Karnataka State Pollution Control Board (KSPCB), ambient air quality data.
3. CPCB, National Ambient Air Quality Standards (NAAQS): PM_{10} 60, $PM_{2.5}$ 40 µg/m^3 (annual).
4. Apache Software Foundation, Apache Spark / PySpark documentation.

08 ACKNOWLEDGEMENTS

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